Hoang Nguyen

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Education

Currently	PhD Candidate in Computer Science (Information Security), ETH Zurich.
9/2021	Master of Science in Computer Science (Formal Methods), Universidad Autonoma de Madrid.
9/2018	Bachelor of Science in Computer Science, Vietnamese-German University.

Employment History

7/2022–Currently	Doctoral assistant, ETH Zurich.
2/2022–6/2022	Research intern, IMDEA Software Institute, Madrid.
9/2021–1/2022	Research assistant, DAAD German Academic Exchange Service.
2/2019–8/2021	Research assistant, Vietnamese-German University.
2/2018–1/2019	Software engineer, Axon Active, Vietnam.

Supervision Activities

Mehdi Berrada	Automated Threat Modeling for ICS	(MS Semester Project, ETH Zurich)
Daniel Gradwohl	Model Transformations for Privacy-by-design	(BS thesis, ETH Zurich)
Karlo Piskor	Secure Object-Relational Mappers	(BS thesis, ETH Zurich)
Thu Le Minh	Fine-grained Access Control in NoSQL	(BS thesis, Vietnamese-German University)
Thu Pham Ho Linh	A web-based validator for data constraints	(BS thesis, Vietnamese-German University)

Teaching Activities

Information Security	Spring 2023, Spring 2024 (TA)	Bachelor's course, ETH Zurich.
Security Engineering	Fall 2022 (TA), Fall 2023 (Head TA)	Master's course, ETH Zurich.

Research Output

1 Research interest

My research focuses on formal methods in software engineering. I apply/propose model-driven methodologies to build secure, reliable software systems.

Fine-grained Access Control on Data-centric Applications In the context of model-driven engineering, [9] discusses some of the challenges and opportunities of the mapping from the Object Contraint Language (OCL) to the Structure Query Language (SQL). [8] defines a novel mapping, called OCL2PSQL , that overcomes some of the aforementioend challenges. [5] proposes the definition of this OCL2PSQL mapping as a case to study the readiness of the state-of-the-art transformation tools. [7] formally characterizes the enforcement of Fine-Grained Access Control (FGAC) policies for a subset of SQL queries. [6] then proposes a model-based methodology for developing an FGAC enforcement mechanism following our formal characterization. [4, 11, 3, 10] extends the methodology on diffrent dimensions, namely, performance, expressiveness, and correctness.

Privacy and Data Protection Data protection is based on both the access control of data and that data is used only for the purposes it was collected. [1] proposes a model-driven development methodology that incorporates privacy policies into the system design.

2 Publications

1. Srdan Krstic, **Hoang Nguyen** and David Basin.
   Model-driven Privacy.
   Privacy Enhancing Technologies Symposium (PETS), 2024.

2. David Basin, Juan Guarnizo, Srdan Krstic, **Hoang Nguyen** and Martin Ochoa.
   Is Modeling Access Control Worth It?
   ACM Conference on Computer and Communications Security (CCS), 2023.

3. **Hoang Nguyen** and Manuel Clavel.
   An extended model-based characterization of fine-grained access control for SQL queries.
   Foundations Track, *Model Driven Engineering Languages and Systems* (MoDELS), 2023.

4. **Hoang Nguyen**.
   Intelligent Enforcement of Fine-Grained Access Control Policies for SQL Queries.
   Master Thesis, Universidad Autonoma de Madrid, 2022.

5. **Hoang Nguyen**, Antonio Garcia-Dominguez and Manuel Clavel.
   The Transformation Tools Contest 2021: OCL2PSQL case.
   Software Technologies: Applications and Foundations (STAF), 3089:1–11, 2021.

6. **Hoang Nguyen** and Manuel Clavel.
   A Model-Driven Approach for Enforcing Fine-Grained Access Control for SQL Queries.
   Springer Nature Computer Science (SNCS), 2(5):370–389, 2020.

7. **Hoang Nguyen** and Manuel Clavel.
   Model-based Characterization of fine-grained Access Control Authorization for SQL Queries.
   Journal of Object Technology (JOT), 19(3):1–13, 2020.

8. **Hoang Nguyen** and Manuel Clavel.
   OCL2PSQL: An OCL-to-SQL Code-Generator for Model-Driven Engineering.
   Future Data and Security Engineering (FDSE), 11814:185–203, 2019.

9. Manuel Clavel and **Hoang Nguyen**.
   Mapping OCL into SQL: Challenges and Opportunities Ahead.
   OCL Workshop, *Model Driven Engineering Languages and Systems* (MoDELS), 2513:3–16, 2019.

3 Drafts

10. **Hoang Nguyen** and Manuel Clavel.
    Proving correctness for SQL implementations of OCL constraints.
    Draft version, arXiv, 2024.

11. **Hoang Nguyen** and Manuel Clavel.
    Optimising Fine-Grained Access Control Policy Enforcement for Database Queries.
    Draft version, arXiv, 2022.

4 Artifacts and Tools

12. Srdan Krstic, **Hoang Nguyen** and David Basin.
    MDP: Model-driven Privacy tools.
    https://polybox.ethz.ch/index.php/s/RoJSS2ECAMww1br

13. **Hoang Nguyen** and Manuel Clavel.
    OCL2PSQL Mapping.
    https://github.com/MoDELSVGU/OCL2PSQL

14. **Hoang Nguyen** and Manuel Clavel.
    SQLSI: SQL Security Injector.
    https://github.com/MoDELSVGU/SQLSI

15. **Hoang Nguyen** and Manuel Clavel.
    SQL2MSFOL: SQL statements to Many-Sorted First-Order Logic.
    https://github.com/MoDELSVGU/SQL2MSFOL